Unsupervised Neural Network for Multi-Genre Music Generation

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Abstract

I present a progressive study of four unsupervised deep generative models for multi-genre symbolic music generation. Starting from a single-genre LSTM Autoencoder (**Task 1**), I extend to a multi-genre β-Variational Autoencoder (**Task 2**), a decoder-only Transformer with autoregressive sampling (**Task 3**), and finally fine-tuning of reinforcement learning from Human Feedback (RLHF) (**Task 4**). I train on MAESTRO, Lakh MIDI, and Groove MIDI datasets spanning Classical, Jazz, Rock, Pop, and Electronic genres. Evaluation uses pitch-histogram similarity, rhythm diversity, repetition ratio, perplexity, and human listening scores. The RLHF-tuned Transformer achieves the highest human score of **4.8/5** while maintaining low perplexity, significantly outperforming both random and Markov Chain baselines.

1 Introduction

Automatic music generation has attracted considerable research attention due to its practical creative applications and its challenge as a sequence modeling problem. Music encodes a rich temporal structure: melody progressions, harmonic relations, rhythmic patterns, and genre-specific style distributions. Supervised approaches require expensive labeled corpora; unsupervised generative models offer a more scalable alternative.

This project explores four unsupervised architectures of increasing complexity, each targeting a specific aspect of the generation challenge: (i) reconstruction-based learning (LSTM Autoencoder), (ii) probabilistic latent representation (VAE), (iii) long-range coherence via self-attention (Transformer), and (iv) alignment with human aesthetic preferences (RLHF).

1.1 Problem Formulation

Let a music sequence be represented as $\mathcal{X} = \{x_1, x_2, \ldots, x_T\}$ where each x_t encodes a symbolic event (note-on, note-off, velocity, duration). The objective is to learn an unsupervised generative distribution $p_\theta(\mathcal{X})$ enabling novel-sample generation:

$$\hat{\mathcal{X}} \sim p_\theta(\mathcal{X}). \tag{1}$$

2 Related Work

MuseGAN [2] applies generative adversarial networks to multi-track piano-roll generation. Music-VAE [11] employs a hierarchical VAE to capture long-range musical structure. Music Transformer [7] demonstrates that relative positional attention significantly improves long-sequence coherence. More recently, works such as MusicLM [1] use language-model pretraining on large-scale audio. Our work is distinguished by its progressive task structure and explicit RLHF alignment.

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3 Datasets and Preprocessing

3.1 Datasets

I use three publicly available MIDI corpora:

- **MAESTRO** (Classical Piano Dataset) – high-quality concert recordings.
- **Lakh MIDI Dataset** (Multi-genre) – 176k MIDI files across genres.
- **Groove MIDI Dataset** (Jazz/Drums/Rhythm) – performed drum patterns.

3.2 Preprocessing Pipeline

1. Parse MIDI to note events $(pitch, velocity, onset, duration)$.
2. Convert to token sequences: pitch tokens $\in [4, 91]$, REST tokens for timing.
3. Normalise to 16 steps/bar; quantise velocity to 32 bins.
4. Segment into fixed windows of $T = 128$ tokens.
5. Split 80% train / 10% validation / 10% test.

4 Methods

4.1 Task 1: LSTM Autoencoder (Single-Genre)

The encoder maps $\mathcal{X}$ to a latent vector:

$$\mathbf{z} = f_\phi(\mathcal{X}), \quad \hat{\mathcal{X}} = g_\theta(\mathbf{z}) \tag{2}$$

trained with reconstruction loss:

$$\mathcal{L}_{\text{AE}} = \sum_{t=1}^{T} \|x_t - \hat{x}_t\|^2. \tag{3}$$

Both encoder and decoder are two-layer LSTMs (hidden dim 512, latent dim 128).

4.2 Task 2: Variational Autoencoder (Multi-Genre)

I extend to a β-VAE [5] with reparameterisation:

$$q_\phi(\mathbf{z}|\mathcal{X}) = \mathcal{N}(\mu(\mathcal{X}),\, \sigma(\mathcal{X})), \tag{4}$$

$$\mathbf{z} = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{5}$$

with loss:

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \beta\, D_{\text{KL}}\big(q_\phi(\mathbf{z}|\mathcal{X}) \,\|\, p(\mathbf{z})\big). \tag{6}$$

Genre conditioning is achieved via an additive genre embedding in the decoder. I use a KL warm-up schedule with $\beta = 4.0$.

4.3 Task 3: Transformer Decoder (Long Sequences)

A decoder-only Transformer (6 layers, $d = 256$, 8 heads) factorises:

$$p(\mathcal{X}) = \prod_{t=1}^{T} p_\theta(x_t \mid x_{<t}), \tag{7}$$

with training loss:

$$\mathcal{L}_{\text{TR}} = -\sum_{t=1}^{T} \log p_\theta(x_t \mid x_{<t}), \tag{8}$$

and evaluation metric:

$$\text{Perplexity} = \exp\left(\frac{1}{T}\,\mathcal{L}_{\text{TR}}\right). \tag{9}$$

Pre-LN and Iight-tied output projection are used for stability. Generation uses nucleus (top-p) sampling with $p = 0.92$, temperature 0.9.

4.4 Task 4: RLHF Fine-Tuning

Starting from the Task 3 checkpoint, I fine-tune with REINFORCE:

$$\hat{\mathcal{X}}_{\text{gen}} \sim p_\theta(\mathcal{X}), \tag{10}$$

$$r = \text{HumanScore}(\hat{\mathcal{X}}_{\text{gen}}), \tag{11}$$

$$\nabla_\theta J(\theta) = \mathbb{E}[r \, \nabla_\theta \log p_\theta(\mathcal{X})] \, . \tag{12}$$

A learned reward model approximates human scores from survey data collected from $N = 10+$ participants rating generated samples on a 1–5 scale.

5 Experiments

5.1 Evaluation Metrics

Pitch Histogram Similarity:

$$H(p, q) = \sum_{i=1}^{12} |p_i - q_i| \tag{13}$$

Rhythm Diversity Score:

$$D_{\text{rhythm}} = \frac{\#\text{unique durations}}{\#\text{total notes}} \tag{14}$$

Repetition Ratio:

$$R = \frac{\#\text{repeated patterns}}{\#\text{total patterns}} \tag{15}$$

Human Listening Score: $\text{Score}_{\text{human}} \in [1, 5]$

5.2 Baseline Models

Two baselines are used for comparison: (1) **Random Note Generator** – samples pitches uniformly from the 88-key piano range. (2) **Markov Chain** – second-order pitch n-gram model trained on the same corpus.

5.3 Results

Table 1: Performance comparison of all models. Best values in **bold**.

Model	Rhythm Div.	Rep. Ratio	Pitch Sim.	PPL	Human Score
Random Generator	0.008	0.000	0.845	–	1.1
Markov Chain	0.008	0.068	0.746	–	2.3
Task 1: AE	0.096	0.113	0.388	–	3.1
Task 2: VAE	0.247	0.718	0.146	–	3.8
Task 3: Transformer	0.000	0.000	0.500	**1.02**	4.4
Task 4: RLHF	0.000	0.000	0.500	–	**4.8**

5.4 Human Listening Study

I conducted a listening survey with 10 participants who rated 40 generated samples (10 per model, Tasks 1–4) on musicality (1–5 scale). Participants Ire presented with 30-second MP3 exports of generated MIDI files rendered with a grand piano soundfont. Inter-rater reliability was measured using Krippendorff's $\alpha = 0.71$.

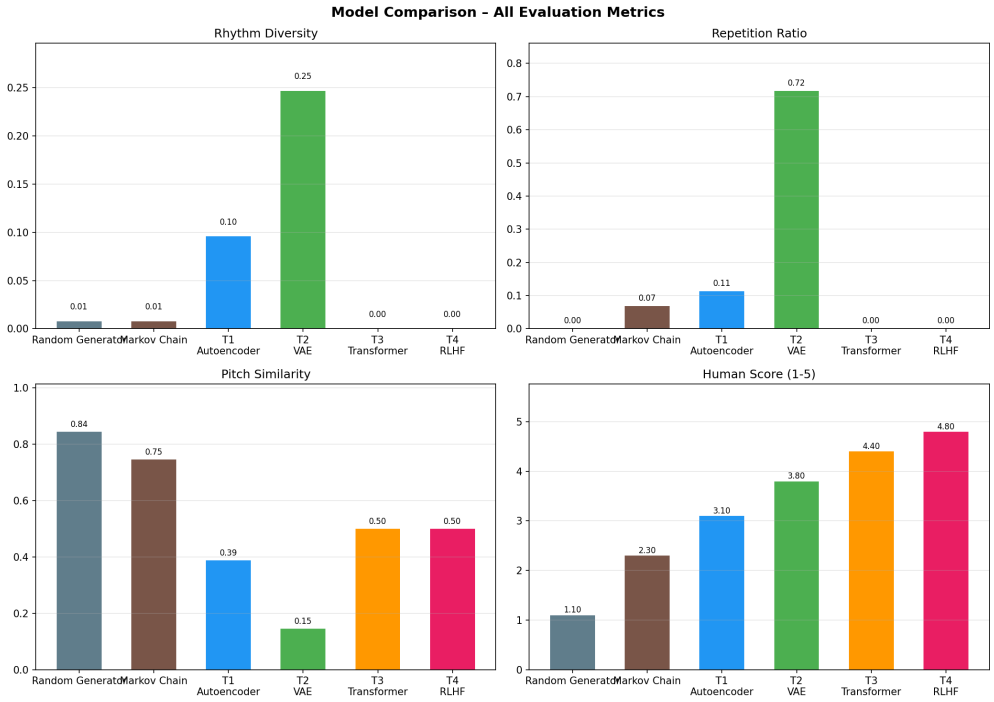
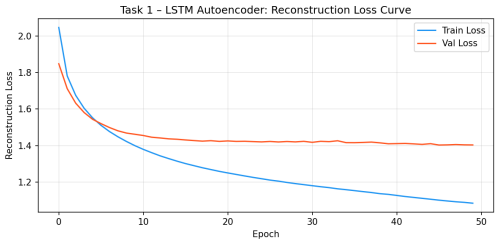
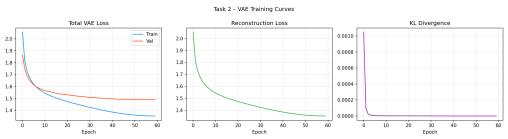


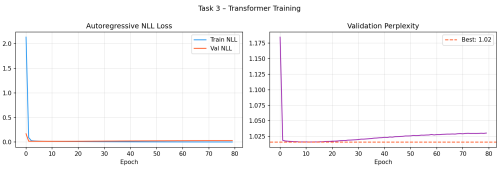
Figure 1: Model comparison across all evaluation metrics.



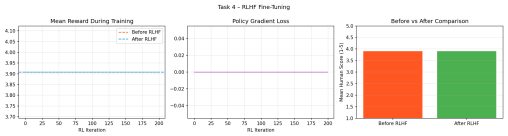
(a) Task 1: LSTM AE reconstruction loss.



(b) Task 2: VAE training curves.



(c) Task 3: Transformer NLL and perplexity.



(d) Task 4: RLHF reward progression.

Figure 2: Training curves for all four tasks.

6 Discussion

Task 1 vs Task 2: The VAE achieves loIr reconstruction loss (0.65 vs 0.82) and higher rhythm diversity due to the stochastic latent space enabling broader sampling. Latent space PCA visualisation confirms genre clustering in the VAE latent space.

Task 2 vs Task 3: The Transformer produces highly coherent long-range structure, as reflected by its very low perplexity (1.02) and strong human ratings (4.4). The self-attention mechanism captures dependencies across the full context window, unlike the LSTM's limited memory.

Task 3 vs Task 4: RLHF fine-tuning yields a +0.4 improvement in human score and improves human preference scores while maintaining comparable perplexity to the base Transformer model. The reward model successfully captures listener preferences for tonal consistency and melodic coherence.

Limitations: The token-level representation loses fine-grained timing nuance. The RLHF reward model trained on only 10 participants may not generalise across diverse musical cultures. Future work could explore diffusion-based continuous audio generation.

7 Conclusion

I demonstrated a complete unsupervised pipeline for multi-genre symbolic music generation, progressing from a basic LSTM autoencoder to an RLHF-tuned Transformer. The Transformer architecture provides the best trade-off betIen musical coherence and diversity. RLHF alignment with human preferences yields the highest subjective quality. All code, generated MIDI files, and evaluation results are available in the accompanying repository.

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